



Explainability vs. Interpretability

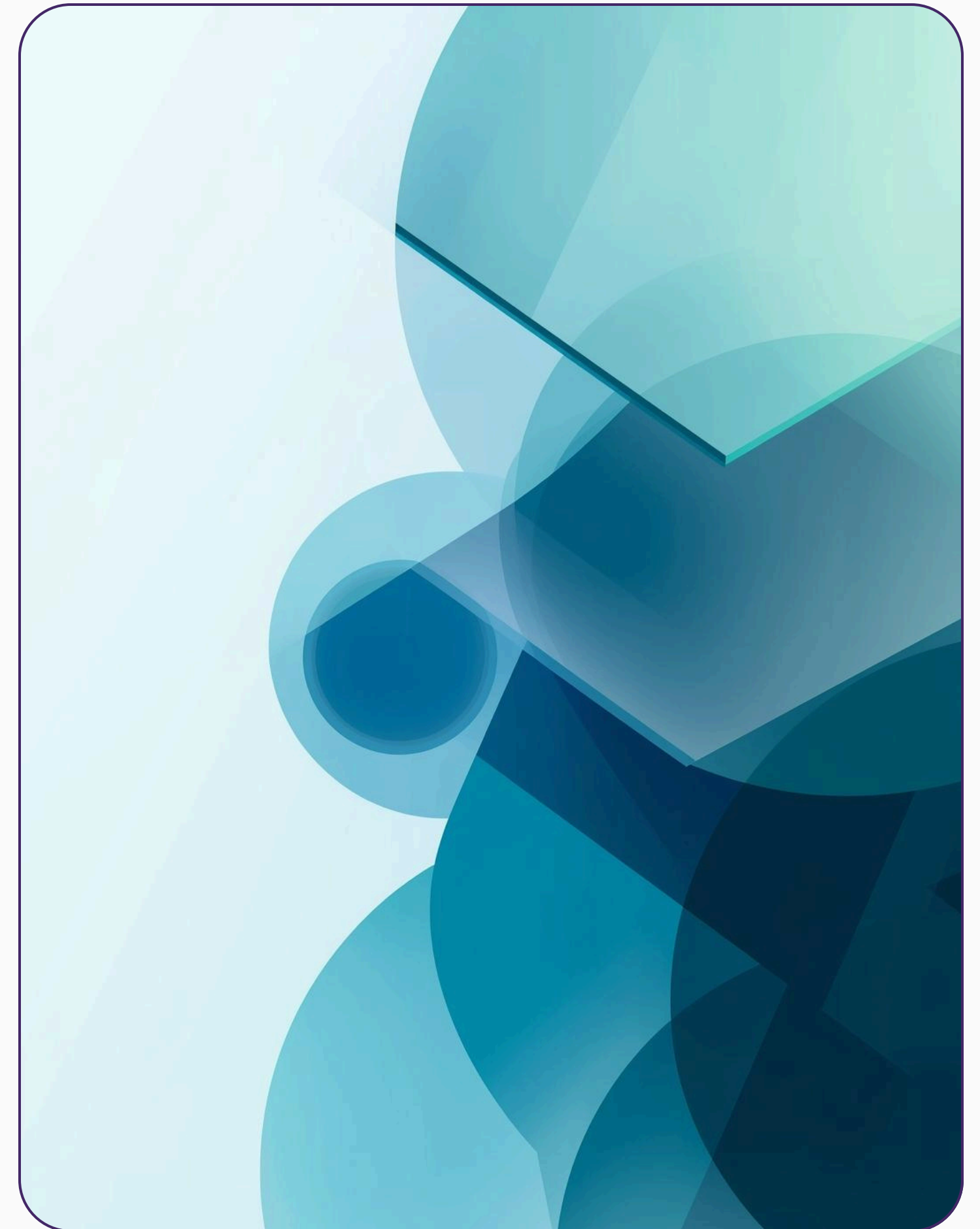


Interpretability

Interpretability refers to how naturally understandable a model is.

- Simple rule-based systems → highly interpretable
- Linear models → moderately interpretable
- Deep neural networks → low interpretability

Interpretability is about structure.

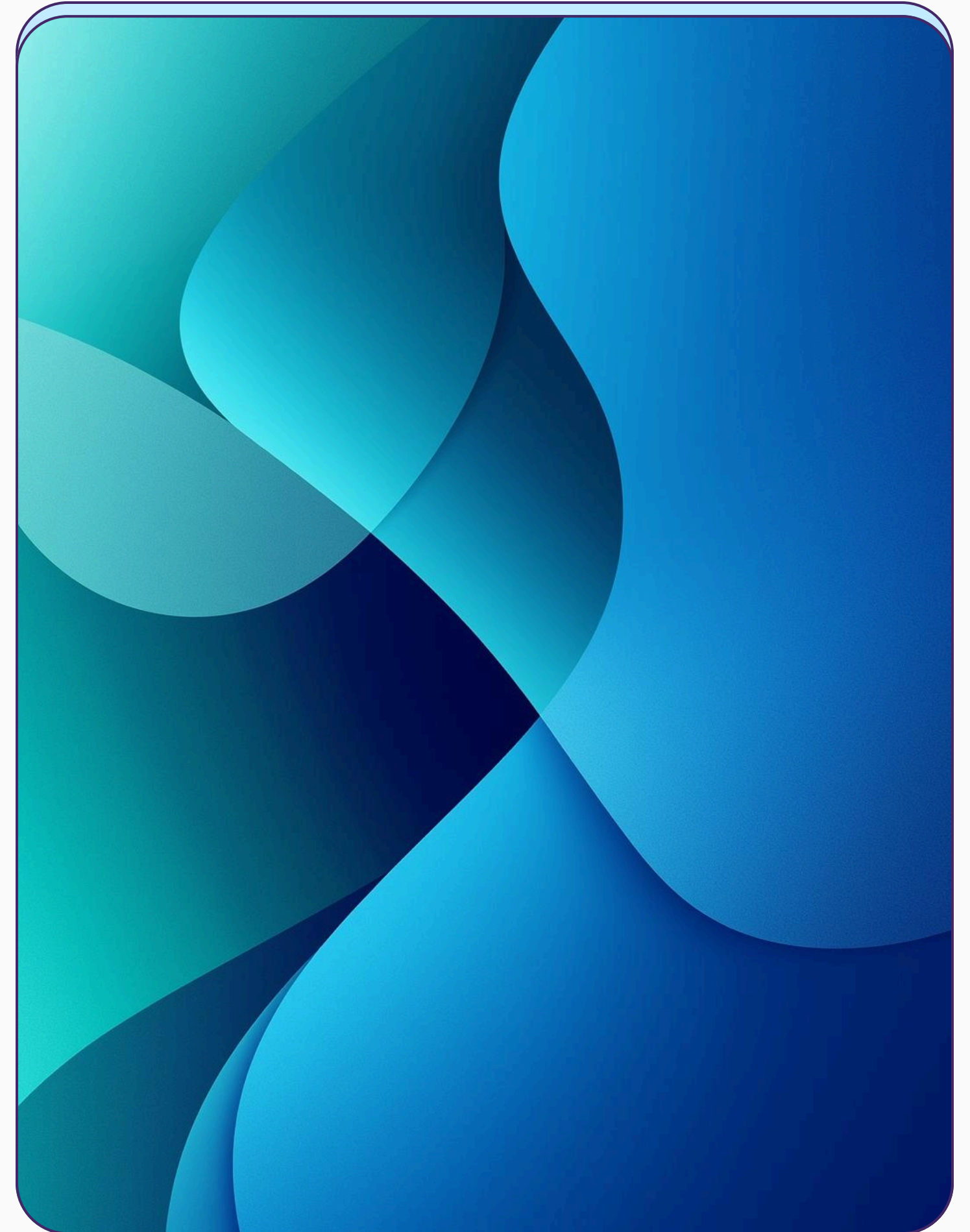


Explainability

Explainability refers to the ability to provide human-understandable reasons for a specific decision.

- Post-hoc explanations
- Feature importance summaries
- Decision rationales presented in plain language

Explainability is about communication, not model simplicity.



Interpretability

- Built into the model
- Structural clarity
- “How it works”

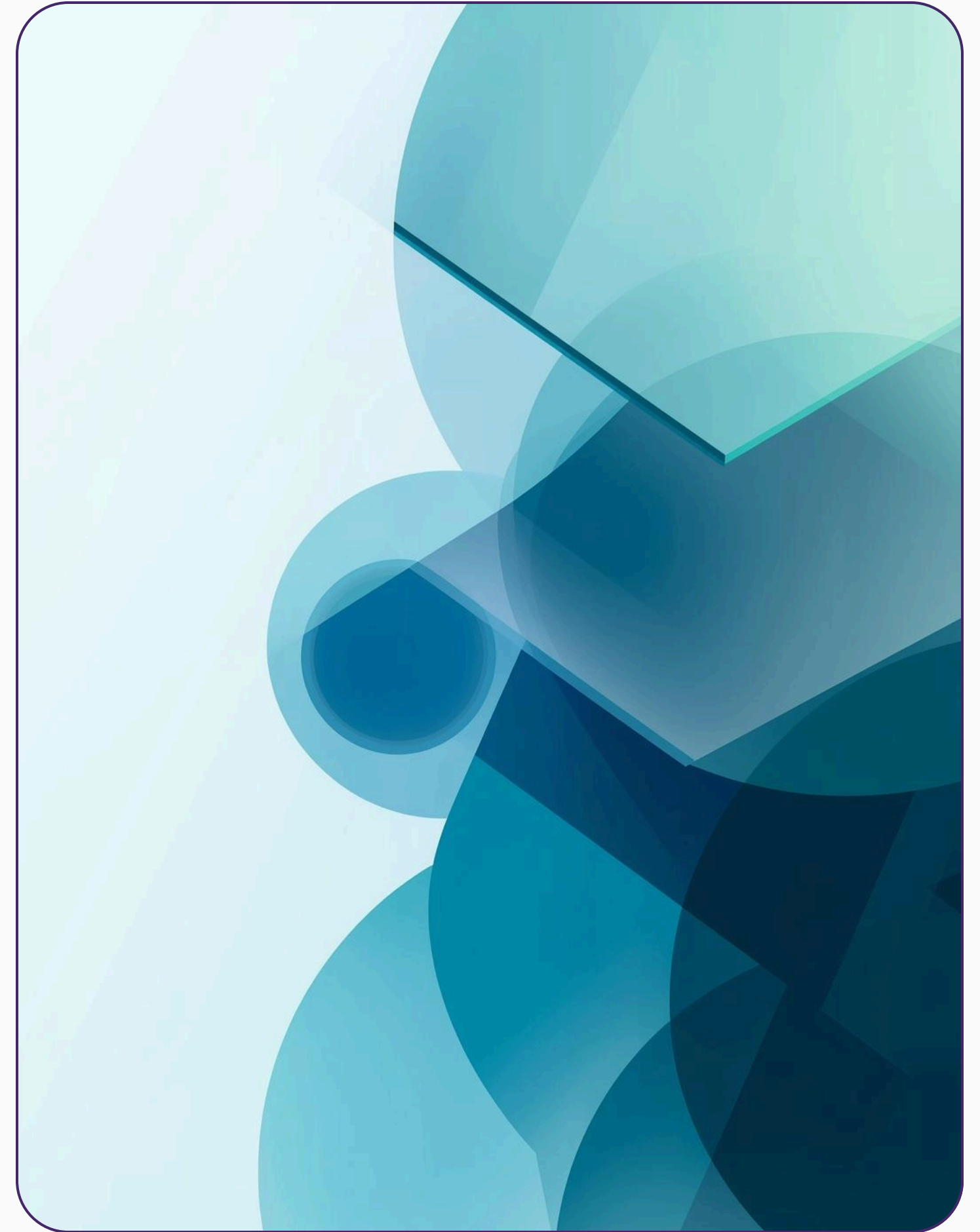
Explainability

- Often added after
- Narrative clarity
- “Why this decision”



The Fundamental Tradeoff

- Most accurate models = least explainable
- Most explainable models = less accurate
- This isn't a technical limitation to solve
- It's inherent in how complex systems work
- Must choose: performance or transparency?



Ethical Insight ...

A system may be accurate but unethical if it cannot provide explanations that affected individuals can understand.

